

Operations Research Lab

Continuous optimization models

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Course outline

Tools

- Introduction and philosophy of the course
- Theory: linear programming
- Theory: nonlinear optimization
- Implementation: linear and nonlinear models
- Advertising budget (convex NLP)
- Continuous location (convex NLP)
- EV charging (LP/QP)
- Queues and capacity (convex NLP)

Deterministic models

- Production and inventory (LP/QP)
- Supply chain and CO₂ (LP/NLP)
- Markowitz portfolio (QP)
- Pricing (NLP)

Uncertainty and machine learning

- Newsvendor (stochastic LP)
- VaR and CVaR (LP)
- Arbitrage and pricing (LP)
- Support Vector Machine (QP)

Lab organization

Introduction

What this lab is about

- Turning **business decisions** into optimization models: how much to produce, where to locate, what price to set, how much risk to accept.
- Implementing the models in **Python + Gurobi**, solving them and — above all — **questioning them**: how much is one extra hour of capacity worth? does the solution hold if the data change by 5%?
- **Continuous variables** only: duality, shadow prices and KKT conditions apply — the tools that turn numbers into economic explanations.

The final question of every lab session

Not just “*what is the optimum?*”, but “*which decision do we recommend, and how robust is it?*”

Notation and abbreviations

- **LP** linear programming; **QP** quadratic; **NLP** nonlinear. *Convex* = every local minimum is global.
- Scalars and indices in lowercase (x_{it}, λ); **vectors** in lowercase bold (x); **matrices** in uppercase bold (Q); **numbered objects** with indices over enumerated sets: $i \in \{1, 2, \dots, n\}$ — the notation of the text is already that of the model; rational data ($\mathbb{Q}$), integer counts ($\mathbb{Z}_{\geq 1}$).
- Dual variables π_i , reduced costs $\bar{c}_j$, slacks $\bar{s}_i$; the **bar** denotes the values of a feasible solution, the **tilde** those of an optimal one ($\tilde{x}_j, \tilde{z}$).
- In every model: first the **data** (with their type), then the **variables**, then objective and constraints explained line by line.

Theory: linear programming

The primal-dual pair (general form)

Data: costs c_j ($j \in N$), coefficients a_{ij} , right-hand sides b_i ($i \in M$). $M_{\leq}, M_{=}, M_{\geq}$ partition M by direction; $N_{\geq 0}, N_{\neq 0}, N_{\leq 0}$ partition N by sign.

$$\begin{aligned} \text{(P)} \quad & \max \sum_{j \in N} c_j x_j \\ \text{subject to} \quad & \sum_{j \in N} a_{ij} x_j \leq b_i, & \forall i \in M_{\leq}, \\ & \sum_{j \in N} a_{ij} x_j = b_i, & \forall i \in M_{=}, \\ & \sum_{j \in N} a_{ij} x_j \geq b_i, & \forall i \in M_{\geq}, \\ & x_j \geq 0, & \forall j \in N_{\geq 0}, \\ & x_j \neq 0, & \forall j \in N_{\neq 0}, \\ & x_j \leq 0, & \forall j \in N_{\leq 0}. \end{aligned}$$

$$\begin{aligned} \text{(D)} \quad & \min \sum_{i \in M} b_i \pi_i \\ \text{subject to} \quad & \sum_{i \in M} a_{ij} \pi_i \geq c_j, & \forall j \in N_{\geq 0}, \\ & \sum_{i \in M} a_{ij} \pi_i = c_j, & \forall j \in N_{\neq 0}, \\ & \sum_{i \in M} a_{ij} \pi_i \leq c_j, & \forall j \in N_{\leq 0}, \\ & \pi_i \geq 0, & \forall i \in M_{\leq}, \\ & \pi_i \neq 0, & \forall i \in M_{=}, \\ & \pi_i \leq 0, & \forall i \in M_{\geq}. \end{aligned}$$

The rules of duality

Primal (max)	Dual (min)
constraint $\leq b_i$ ($i \in M_{\leq}$)	variable $\pi_i \geq 0$
constraint $= b_i$ ($i \in M_{=}$)	variable $\pi_i \begin{smallmatrix} \geq \\ \leq \end{smallmatrix} 0$
constraint $\geq b_i$ ($i \in M_{\geq}$)	variable $\pi_i \leq 0$
variable $x_j \geq 0$ ($j \in N_{\geq 0}$)	constraint $\geq c_j$
variable $x_j \begin{smallmatrix} \geq \\ \leq \end{smallmatrix} 0$ ($j \in N_{\begin{smallmatrix} \geq \\ \leq \end{smallmatrix} 0}$)	constraint $= c_j$
variable $x_j \leq 0$ ($j \in N_{\leq 0}$)	constraint $\leq c_j$

- **Equalities** in the primal give **free** dual variables (and vice versa).
- For a *minimization* primal the table is read from right to left.

The two duality theorems

Weak duality

For every $(\bar{x}_1, \dots, \bar{x}_n)$ feasible for (P) and every $(\bar{\pi}_1, \dots, \bar{\pi}_m)$ feasible for (D):

$$\sum_{j \in N} c_j \bar{x}_j \leq \sum_{i \in M} b_i \bar{\pi}_i$$

(any feasible dual is an **upper bound** for the primal).

Strong duality

(P) has an optimum $(\tilde{x}_1, \dots, \tilde{x}_n) \iff$ (D) has an optimum $(\tilde{\pi}_1, \dots, \tilde{\pi}_m)$, and then the gap closes:

$$\sum_{j \in N} c_j \tilde{x}_j = \sum_{i \in M} b_i \tilde{\pi}_i.$$

Building the dual with the rules: the 2×2 LP

Canonical maximization form: every $\leq$ constraint gives a $\pi_i \geq 0$; every variable ≥ 0 gives a constraint $\geq c_j$ (coefficients read by **column**); the objective swaps roles.

$$\begin{array}{ll} \max & 30x_1 + 50x_2 \\ \text{subject to} & x_1 + 3x_2 \leq 90, \\ & 2x_1 + x_2 \leq 80, \\ & x_1, x_2 \geq 0. \end{array}$$

$$\begin{array}{ll} \min & 90\pi_1 + 80\pi_2 \\ \text{subject to} & \pi_1 + 2\pi_2 \geq 30, \\ & 3\pi_1 + \pi_2 \geq 50, \\ & \pi_1, \pi_2 \geq 0. \end{array}$$

Building the dual: all cases in a single LP

A *minimization* problem with three directions and three signs (table read from right to left):

$$\begin{array}{llll}
 \min & 5x_1 + 8x_2 - 9x_3 & & \\
 \text{subj. to} & x_1 + x_2 & \geq & 30, \\
 & x_1 + x_2 - x_3 & = & 100, \\
 & x_1 - 2x_2 & \leq & -20, \\
 & x_1 & \geq & 0, \\
 & x_2 & \geq & 0, \\
 & x_3 & \leq & 0.
 \end{array}$$

$$\begin{array}{llll}
 \max & 30\pi_1 + 100\pi_2 - 20\pi_3 & & \\
 \text{subj. to} & \pi_1 + \pi_2 + \pi_3 & \leq & 5, \\
 & \pi_1 + \pi_2 - 2\pi_3 & = & 8, \\
 & -\pi_2 & \geq & -9, \\
 & \pi_1 & \geq & 0, \\
 & \pi_2 & \geq & 0, \\
 & \pi_3 & \leq & 0.
 \end{array}$$

The LP optimality conditions: complementary slackness

Unknown variables plain; bar = values of a **feasible** solution; tilde = of an **optimal** one. Slack and reduced cost (the slack of the *dual* constraint; the bar in $\bar{c}_j$ is part of the name):

$$s_i = \sum_{j \in N} a_{ij} x_j - b_i, \quad \bar{c}_j = c_j - \sum_{i \in M} a_{ij} \pi_i.$$

Theorem (complementary slackness)

Feasible $(\bar{x}_1, \dots, \bar{x}_n)$, $(\bar{\pi}_1, \dots, \bar{\pi}_m)$ are **optimal** for (P) and (D) $\iff$

$$\bar{\pi}_i \cdot \bar{s}_i = 0 \quad \forall i \in M, \quad \bar{x}_j \cdot \bar{c}_j = 0 \quad \forall j \in N.$$

- **Slack constraint $\Rightarrow$ zero dual**; variable $\neq 0 \Rightarrow$ zero reduced cost (implications: degeneracy).

The signs of the shadow prices

One convention: $\tilde{\pi}_i = \frac{\partial \tilde{z}}{\partial b_i}$, valid for $b_i \in [b_i^{\min}, b_i^{\max}]$ (validity range)

- Two questions: does increasing b_i **enlarge** ($\leq$) or **shrink** ($\geq$) the feasible region? And a larger region can never worsen the optimum.

Constraint direction	minimization	maximization
$\leq$ ($b_i \uparrow \Rightarrow$ larger region)	$\tilde{\pi}_i \leq 0$	$\tilde{\pi}_i \geq 0$
$\geq$ ($b_i \uparrow \Rightarrow$ smaller region)	$\tilde{\pi}_i \geq 0$	$\tilde{\pi}_i \leq 0$
$=$	any sign (free dual)	

- In a minimization: $\leq$ (capacity) $\Rightarrow$ loosening reduces the cost, $|\tilde{\pi}_i|$ = marginal saving; $\geq$ (requirement) $\Rightarrow \tilde{\pi}_i$ = marginal cost. Non-binding constraint: $\tilde{\pi}_i = 0$.

Duality on the 2×2 LP: solving gives, we verify

1. **Solving gives:** $(\tilde{x}_1, \tilde{x}_2) = (30, 20)$, $\tilde{z} = 1900$, $(\tilde{\pi}_1, \tilde{\pi}_2) = (14, 8)$, $(\bar{c}_1, \bar{c}_2) = (0, 0)$; both constraints binding.
2. Check (complementary slackness): $x_1, x_2 \neq 0 \Rightarrow$ dual constraints binding:
 $14 + 16 = 30 = c_1$, $42 + 8 = 50 = c_2$. ✓
3. Check of strong duality: $90 \cdot 14 + 80 \cdot 8 = 1900 = \tilde{z}$. ✓
4. Reduced costs $\bar{c}_j = c_j - \sum_i a_{ij} \tilde{\pi}_i$: $\bar{c}_1 = 30 - (14 + 16) = 0$, $\bar{c}_2 = 50 - (42 + 8) = 0$
(zero: both variables are nonzero — beware of **degeneracy**: there can also be variables at zero with $\bar{c} = 0$).

Sensitivity on the 2×2 example: shadow prices and reduced costs

Shadow prices

Increase b_1 by one unit at a price of 12? Worth it ($14 > 12$). With $b_1 = 100$:
 $2040 = 1900 + 10 \cdot 14$ — exact as long as b_1 stays in the validity range $[b_1^{\min}, b_1^{\max}] = [40, 240]$.
A non-binding constraint always has zero shadow price.

Reduced costs: is a third variable worth it?

Coefficient $c_3 = 20$, consumptions $a_{13} = a_{23} = 1$:

- at the shadow prices the resources are worth $1 \cdot 14 + 1 \cdot 8 = 22 > 20$;
- solving: $\tilde{x}_3 = 0$ with $\bar{c}_3 = 20 - 22 = -2$ and $c_3^{\max} = 22 \Rightarrow$ worth it only from $c_3 = 22$ upwards;
- cross-check with $c_3 = 23$: the solution changes to $(0, 5, 75)$, value 1975;
- validity ranges of the coefficients: c_1 may vary in $[16.7, 100]$ without changing the solution.

All cases in a single LP: checking the conditions

- **Solving gives:** $(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = (60, 40, 0)$, $\tilde{z} = 620$; $(\tilde{\pi}_1, \tilde{\pi}_2, \tilde{\pi}_3) = (0, 6, -1)$; $(\bar{c}_1, \bar{c}_2, \bar{c}_3) = (0, 0, -3)$; coefficient ranges $(-\infty, 8]$, $[5, 17]$, $(-\infty, -6]$.
- Checks: $\tilde{\pi}_1 = 0$ (slack constraint: $100 > 30$); $\tilde{\pi}_2 = 6$ (equality, free dual); $\tilde{\pi}_3 = -1 \leq 0$ ($\leq$ in a minimization); strong duality $600 + 20 = 620$. ✓
- x_3 at its **upper bound** (zero): $\bar{c}_3 = -9 + 6 = -3$, threshold $c_3^{\max} = -6$; perturbations: $b_2+1 \rightarrow 626$, $b_3+1 \rightarrow 619$, $x_3 = -1 \rightarrow 623$. ✓

Theory: nonlinear optimization

A 2×2 QP, worked out in full

Separable quadratic objective, one threshold constraint on the sum:

$$\begin{array}{ll} \min & x_1^2 + 2x_2^2 \\ \text{subject to} & x_1 + x_2 \geq 6, \\ & x_1, x_2 \geq 0. \end{array}$$

- $Q = \text{diag}(2, 4) \succ 0$: convex $\Rightarrow$ certified global optimum.
- **Solving gives:** $(\tilde{x}_1, \tilde{x}_2) = (4, 2)$, $\tilde{f} = 24$, $\lambda = 8$.
- Check: binding constraint ($4 + 2 = 6$) and **equal derivatives** $2\tilde{x}_1 = 8 = 4\tilde{x}_2 =$ shadow price of the threshold. ✓

KKT conditions

For $\min f(x_1, \dots, x_n)$ subject to $g_i(x_1, \dots, x_n) \leq 0$, $h_j(x_1, \dots, x_n) = 0$, with Lagrangian $L = f + \sum_i \lambda_i g_i + \sum_j \nu_j h_j$ ($\lambda_i \geq 0$), at a regular optimum $(\tilde{x}_1, \dots, \tilde{x}_n)$:

$$\frac{\partial L}{\partial x_k}(\tilde{x}_1, \dots, \tilde{x}_n) = 0, \quad \forall k \in \{1, 2, \dots, n\} \quad (\text{stationarity})$$

$$\lambda_i g_i(\tilde{x}_1, \dots, \tilde{x}_n) = 0, \quad \forall i \in \{1, 2, \dots, m\} \quad (\text{complementarity})$$

If the problem is convex, they are also **sufficient**.

Worked example: KKT on the example QP

$\min x_1^2 + 2x_2^2$ subject to $x_1 + x_2 \geq 6$. Stationarity: $2x_1 = \lambda$, $4x_2 = \lambda$; the constraint is binding, so $\lambda = 8$, $(\tilde{x}_1, \tilde{x}_2) = (4, 2)$, $\tilde{f} = 24$; with $6 + \varepsilon$: $\tilde{f} = 24 + 8\varepsilon + \frac{2}{3}\varepsilon^2$.

- In the linear case the KKT conditions **are** the complementary slackness of (P)–(D): multipliers = duals π_i .
- No duals available: estimate them **by perturbation**.

The sensitivity analysis protocol (in every lab session)

1. **Base scenario:** solve, verify, binding constraints.
2. **One-at-a-time:** one key parameter over a grid.
3. **Shadow prices and reduced costs:** dual vs perturbed re-optimization; $\bar{c}_j$ of the variables at zero = break-even threshold.
4. **Scenarios:** pessimistic, central, optimiztic.
5. **Trade-off:** a frontier (cost-service, risk-return, cost-CO₂).
6. **Stability:** data $\pm 5\%$.

Implementation: linear models

Building a model in five steps

```
m = gp.Model("production")           # 1. container
x = m.addVars(products, months, name="x") # 2. variables (>= 0)
m.addConstrs((x.sum("*", t) <= b[t]      # 3. constraints
              for t in months), name="cap")
m.setObjective(..., GRB.MINIMIZE)      # 4. objective
m.write("model.lp"); m.optimize()      # 5. debug + solution
```

- **Free** variables (SVM: b ; CVaR: η): `m.addVar(lb=-GRB.INFINITY)` — forgetting it is the most common error!
- Sums: `gp.quicksum(...)` or `x.sum(i, "*")`.

Reading the solution

```
if m.Status == GRB.OPTIMAL:          # ALWAYS before reading numbers
    print(m.ObjVal, x[i,t].X)
elif m.Status == GRB.INFEASIBLE:
    m.computeIIS(); m.write("conflict.ilp")
```

Pi	shadow price of the constraint (LP)
Slack	slack: 0 $\Rightarrow$ binding constraint
RC	reduced cost of a variable at zero
SARHS _{Low/Up}	validity range of the shadow price
SAObj _{Low/Up}	validity range of the reduced cost

- Minimization problems, $\leq$ constraint: $P_i \leq 0$ (Gurobi convention).
- RHS with constants on the left: Gurobi moves them into the right-hand side (**pitfall** in sensitivity analysis — see EV charging).

Interpretation checklist

1. Status OPTIMAL? If not: diagnose (IIS, .lp file).
2. Is the order of magnitude of the optimum sensible?
3. Binding constraints = the expected ones?
4. Shadow prices: which resource to expand first, and up to what price?
5. Stability: data $\pm 5\%$.
6. Managerial recommendation in three lines.

Implementation: nonlinear models

One solver only: general NLPs in Gurobi too

- Nonlinear functions as **function constraints** on auxiliary variables:
`addGenConstrLog/Exp/Pow` with `FuncNonlinear=1`.
- Bilinear terms and ratios with an auxiliary variable and `NonConvex=2` (e.g. $w(\mu - \lambda) = 1$).
- **Certified global** optimum, same interpretation checklist.
- For marginal analyses: `MIPGap`, `FeasibilityTol`, `OptimalityTol` at 10^{-9} ; reformulate for scaling (e.g. $q \cdot p^\varepsilon \leq A$, not $q \leq A p^{-\varepsilon}$).

Multi-period production and inventory (LP/QP)

Production and inventory — the problem

- **Decisions:** production x_{it} and inventory s_{it} (nonnegative); **objective:** minimum cost.
- **Data:** demand d_{it} , costs c_{it}, h_{it} , hours per unit a_i , hours available b_t , initial inventory s_{i0} .

Model (LP)

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{t=1}^m (c_{it} x_{it} + h_{it} s_{it}) \\ \text{subject to} \quad & s_{i,t-1} + x_{it} = d_{it} + s_{it}, & \forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}, \\ & \sum_{i=1}^n a_i x_{it} \leq b_t, & \forall t \in \{1, 2, \dots, m\}, \\ & x_{it}, s_{it} \geq 0, & \forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}. \end{aligned}$$

Production and inventory — understanding the constraints

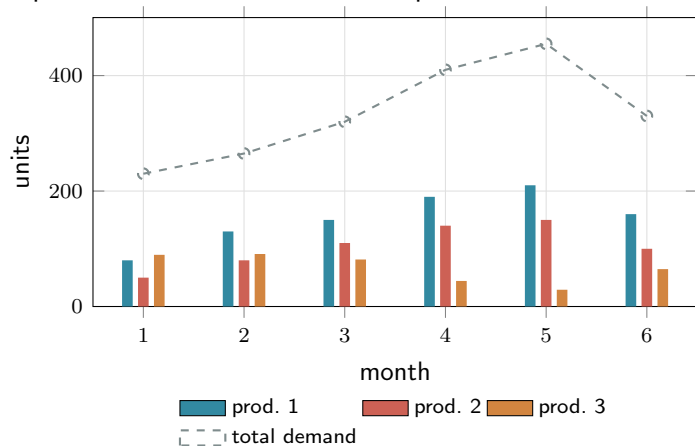
- **Objective:** the model freely trades early production (more inventory) against production at the capacity limit.
- **Balance:** inventory accounting; it is the constraint that **links the months**: without it the problem splits into $|T|$ independent problems.
- **Capacity:** the shared scarce resource; its dual says how much one extra hour is worth.

Worked example by hand (1 product, 2 months)

$d = (60, 100)$, capacity 80/month, $c = 10$, $h = 1$: the only feasible plan is $x = (80, 80)$, $s_1 = 20$, cost 1.620 €. +1 hour in month 2 $\Rightarrow$ cost -1 € (one month of holding less): dual of month 2 = -1 , month 1 = 0.

Production and inventory — case study

3 products $\times$ 6 months, demand peak in months 4–5, 420–460 hours/month.



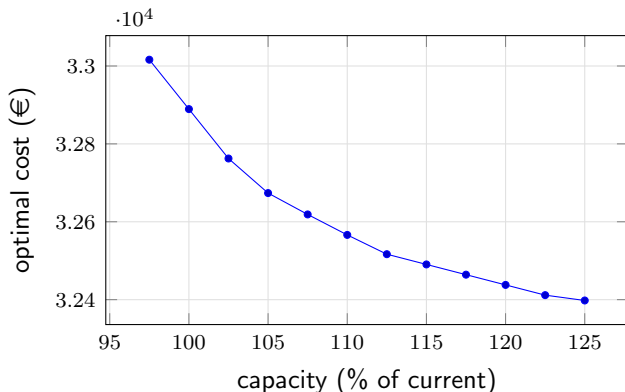
- Optimal cost: **32.889 €**
- Product 3 (2.1 hours/unit) is produced **early** (*pre-build*) and stocked (up to 107 units) to free up the peak months.
- Zero inventory in the last month: the model does not waste.

Production and inventory — the shadow prices tell a story

month	1	2	3	4	5	6
hours used	330/420	420/420	460/460	460/460	460/460	420/420
shadow price (€/h)	0	-0.76	-1.52	-2.29	-3.05	-3.81

- The duals grow by $0.762 = h_3/a_3$ per month: one extra hour in month t saves **one month of holding** product 3.
- Check by perturbation: +1 hour in month 6 $\Rightarrow$ cost -3.810 € (it matches the dual).
- When the solver and economic reasoning give the same number, the model is probably right.

Production and inventory — smoothing and the value of capacity



- **Convex and decreasing** curve: the first extra hours are worth a lot.
- Below 97%: **infeasible** — demand can no longer be met.
- QP variant with $\gamma \sum_{t>1} (X_t - X_{t-1})^2$: an almost flat profile (max change from 81 down to 1 unit) for +114 € (+0.35%).

Take-away

Overtime in months 5–6 pays off up to 3–3.8 €/hour below the internal price; the levelled plan is almost free.

Supply chain with congestion and CO₂ (LP/NLP)

Supply chain — the problem

In words. Decide how many units to ship on each leg of the network. Objective: minimum transport cost. Constraints: flow conservation at every node and capacity of the legs.

Model (minimum cost flow)

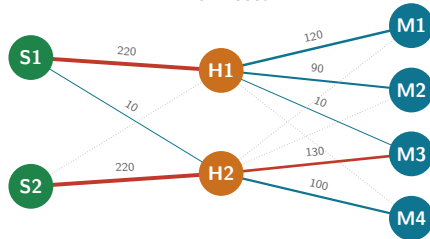
$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad \text{subject to} \quad \sum_{j: (i,j) \in A} x_{ij} - \sum_{j: (j,i) \in A} x_{ji} = b_i, \quad \forall i \in N, \quad 0 \leq x_{ij} \leq u_{ij}$$

Objective variants: quadratic congestion $\sum (c_{ij} x_{ij} + \alpha c_{ij} x_{ij}^2 / u_{ij})$ (convex); internal CO₂ price: $\sum (c_{ij} + \tau e_{ij}) x_{ij}$; minimax of the utilization.

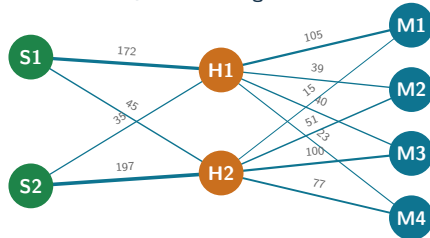
- Worked example (2 routes): the LP saturates the cheap route; congestion equalises the **marginal costs** and leaves a safety margin.

Supply chain — case study

Minimum-cost LP



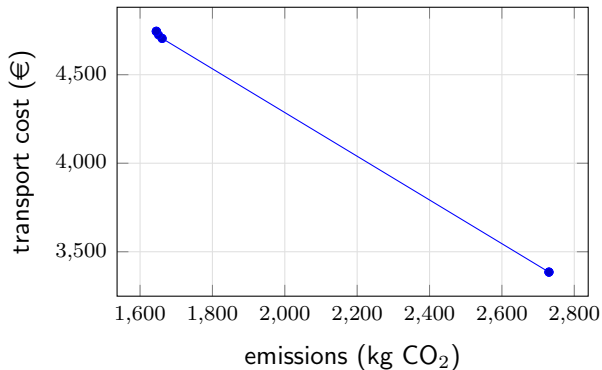
Quadratic congestion



2 plants → 2 hubs → 4 markets; cheap “road” legs (polluting), expensive “rail” ones (clean).

- LP: cost 3.385 €, 3 saturated arcs (in red), emissions 2.730 kg.
- Congestion: cost 3.703 €, no arc above 90%.
- Shadow prices of demand = marginal costs of serving the markets: M1 8.00 — M4 10.50 €/unit.

Supply chain — the price of CO₂



- $\tau \leq 1$: “road” configuration: 3.385 € and 2.730 kg.
- $\tau \geq 1.5$: “rail” configuration: 4.705 € and 1.661 kg.
- Exact threshold, by bisection: $\tau^* = 1.25$ €/kg.
- LP solutions sit at the **vertices**: the frontier moves in jumps; below the threshold the internal price changes nothing.

Warning

Pure minimax ($\min z$) is degenerate: every routing with utilization $\leq z^*$ is optimal, however expensive. Always combine it with the cost.

Markowitz portfolio (QP)

Markowitz — the problem

In words. Split the capital across n assets. Objective: minimum variance. Constraints: weights summing to 1, minimum expected return, per-asset limits.

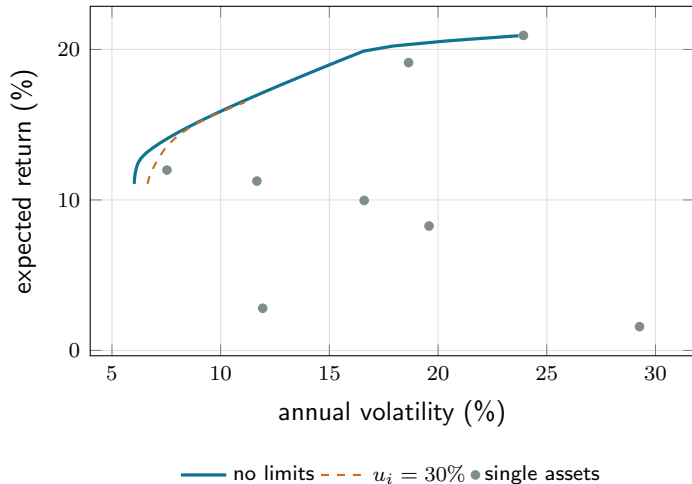
Model (convex QP: $Q \succeq 0$ always)

$$\min \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j \quad \text{subject to} \quad \sum_{i=1}^n \mu_i x_i \geq \bar{r}, \quad \sum_{i=1}^n x_i = 1, \quad \ell_i \leq x_i \leq u_i$$

Worked example (2 uncorrelated assets)

$\sigma_1 = 20\%$, $\sigma_2 = 30\%$: $x_1 = 9/13 = 69.2\%$, portfolio volatility 16.6% — **less than either asset**: diversification in its purest form.

Markowitz — the efficient frontier



8 ETFs, μ and Q **estimated** from 60 months.

- Minimum variance: return 11%, volatility **6%** < 7.5% of the best single asset.
- Equally weighted $1/n$: **dominated** (10.7%, 10.1%).
- Cap $u_i = 30\%$: it cuts the top — the cost of the mandates *is visible*.

Markowitz — sensitivity and fragility

- The return constraint is **non-binding** up to $\bar{r} = 11.06\%$ (minimum variance already yields a lot): asking for “at least 8%” costs nothing.
- Beyond that: every point of return is paid for in variance (multiplier ≈ 0.022 at $\bar{r} = 12\%$).
- Composition along the frontier: from the defensive mix (UTL, CON, SAN) to a portfolio concentrated in the aggressive assets (MAT, ENE).

The most important practical lesson

Return estimates are **far more unstable** than covariance estimates (standard error $\approx 2.6\%$ per year over 60 months): optimized portfolios chase estimation errors. Remedies: u_i limits, shrinkage, minimum variance.

Pricing and revenue management (NLP)

Pricing — the problem

In words. Decide price p and quantity q . Objective: maximum profit $(p - c)q$. Constraints: $q \leq d(p)$ and $q \leq k$ (capacity).

Model and demand curves

$$\max p q - c q \text{ subject to } q \leq d(p), q \leq k$$

linear $d(p) = a - bp$; constant elasticity $d(p) = \theta p^{-\varepsilon}$; logistic $d(p) = m/(1 + e^{-\alpha + \beta p})$.

- The nonconvexity lies in the product $p \cdot q$ (bilinear); with linear demand the reduced profit is a concave parabola; Gurobi still solves it to global optimality (NonConvex=2).

Pricing — worked example and case study

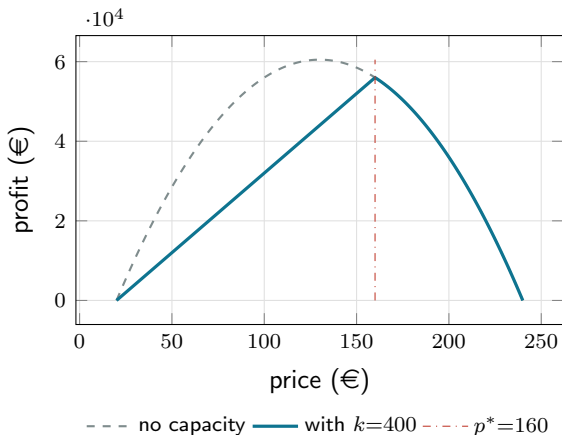
Concert: $d(p) = 1200 - 5p$, $c = 20$ €, $k = 400$ seats.

1. Without capacity: $p^o = (a/b + c)/2 = 130$ €, $q^o = 550$.
2. Capacity bites: $p^* = (a - k)/b = \mathbf{160}$ €, profit 56.000 €.
3. Value of one extra seat: $(p^* - c) + k dp^*/dk = 140 - 80 = \mathbf{60}$ € — **not** the full margin: filling the seat means lowering the price for everyone. (Gurobi, by perturbation: 59.80.)

The optimum at the kink

With constant elasticity the unconstrained optimum would be 36.67 €, but there demand explodes: the true optimum is at the kink $d(p) = k$, i.e. $p^* = (\theta/k)^{1/\varepsilon} = 79.11$ € — no derivative vanishes.

Pricing — profit and seating capacity



- The capacity constraint cuts the parabola and moves the optimum from 130 to 160 €.
- The value-of-capacity curve flattens at $q^o = 550$: beyond that, seats are worth nothing.
- Multi-product (stalls/balcony with substitution): prices to be set **jointly**; profit 85.149 €.
- Expansions: the balcony pays (+3.838 € for 50 seats), the stalls do not (+3.287 €).

Advertising budget (convex NLP)

Advertising budget — the problem and the KKT condition

In words. Split the budget b across channels with concave response $r_i(x_i) = a_i \log(1 + k_i x_i)$; constraints: budget and caps u_i .

$$\max \sum_{i=1}^n r_i(x_i) \quad \text{subject to} \quad \sum_{i=1}^n x_i \leq b, \quad 0 \leq x_i \leq u_i$$

KKT = economics

At the optimum $r'_i(x_i^*) = \lambda$ for every interior channel: **the last euro returns the same in every active channel**. Channels at their cap: marginal $> \lambda$; channels at zero: initial marginal $< \lambda$.

Worked example ($b = 50$): $x = (35.6; 14.4)$, $\lambda = 2.46$.

Advertising budget — numerical check and value curve

	social	search	TV	influencer
spend (thousand €)	24.3	34.8	22.9	18.0
marginal return	8.2693	8.2693	8.2693	8.2693

- Check: +1.000 € of budget $\Rightarrow$ response $+8.244 \approx \lambda$.
- TV gets less than social despite the higher cap: what matters is the **saturation speed** k_i , not the size of the channel.
- λ falls from 19 ($b = 20$) to 0 ($b = 300$, all at their caps): the λ curve is the argument for negotiating the budget with the CFO.

Continuous location (convex NLP)

Location — three objectives, three answers

In words. Decide the coordinates (x, y) of the facility; the objective — average, maximum or quadratic — is the real managerial choice.

$$\text{Weber: } \min \sum_{i=1}^n w_i \sqrt{(x - a_i)^2 + (y - b_i)^2}$$

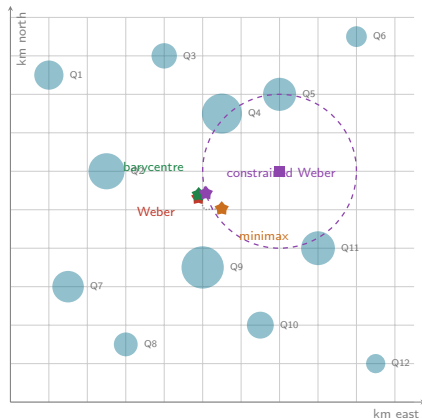
$$\text{Minimax: } \min z \quad \text{subject to} \quad \sqrt{(x - a_i)^2 + (y - b_i)^2} \leq z, \quad \forall i \in \{1, 2, \dots, n\}$$

Worked example in 1D (weights 1, 1, 3 at km 0, 4, 10)

Weber = weighted median = 10; centroid = 6.8; minimax = 5. Three reasonable objectives, three different answers.

- Quadratic $\Rightarrow$ weighted centroid (closed form); Weber: not differentiable at the demand points; minimax: centre of the smallest enclosing circle, **it ignores the weights**.

Location — case study and equity



12 districts; constraint: within 2 km of the substation at (7, 6).

- Weber (4.89; 5.31): average 3.344 km, max 6.314.
- Minimax (5.50; 5.03): average 3.391, max **5.680**.
- Substation constraint: +0.2% — **almost free**.
- The efficiency-equity frontier (ϵ -constraint) is almost flat: -0.63 km for the farthest district costs 47 m of average.

EV charging (LP/QP)

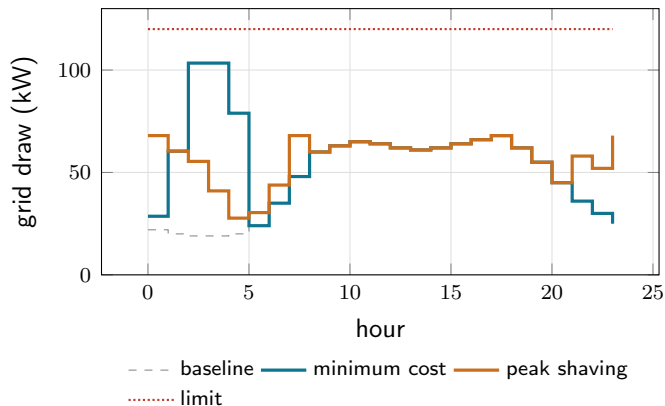
EV charging — the problem

In words. How much power x_{vt} per vehicle and hour, at minimum cost; availability $a_{vt} \in \{0,1\}$ is **data**, not a decision.

Model (LP: no binary variables!)

$$\begin{aligned} \min \quad & \sum_{v=1}^n \sum_{t=1}^m \pi_t \Delta t x_{vt} \\ \text{subject to} \quad & \eta \sum_{t=1}^m \Delta t x_{vt} \geq e_v, & \forall v \in \{1, 2, \dots, n\}, \\ & \sum_{v=1}^n x_{vt} + b_t \leq k, & \forall t \in \{1, 2, \dots, m\}, \\ & 0 \leq x_{vt} \leq a_{vt} \bar{p}_v, & \forall v \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}. \end{aligned}$$

EV charging — minimum cost vs peak shaving



- Minimum cost: 20.36 €/night, peak **103 kW**.
- Peak shaving: peak 68 kW; with the cost+ ρ peak compromise the same peak costs 22.15 € (pure minimax: 27.79!).
- Duals of the energy needs = $0.08/\eta$ or $0.09/\eta$: the price of each vehicle's **marginal hour**.
- Minimum feasible capacity: 68 kW (bisection).

A real pitfall

In $\text{quicksum}(x) + \text{base}[t] \leq C$ the stored RHS is $C - \text{base}[t]$.

Queues and service capacity (convex NLP)

M/M/1 queues — the problem

In words. How much service capacity to buy; objective: capacity cost + value of the customers' time; constraint: stability $\mu > \lambda$.

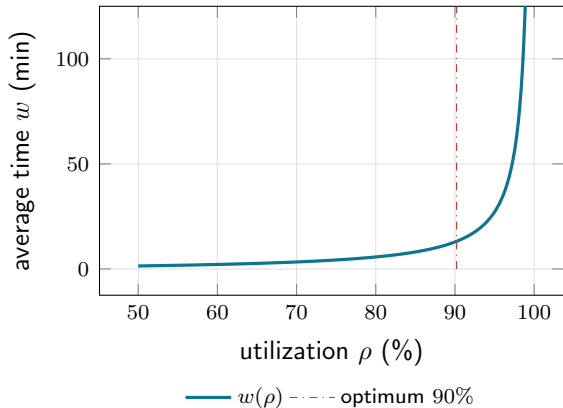
Model and closed-form solution

$$\min_{\mu > \lambda} c\mu + h \frac{\lambda}{\mu - \lambda} \quad \Longrightarrow \quad \mu^* = \lambda + \sqrt{h\lambda/c}$$

“capacity = demand + **buffer**”; the buffer grows like $\sqrt{\lambda}$ (economies of scale, pooling).

Worked example ($\lambda = 8$, $c = 2$, $h = 4$): $\mu^* = 12$, $\rho = 67\%$, cost 32 €/h; with $\mu = 9$ (“more efficient”, $\rho = 89\%$): 50 €/h.

Queues — the utilization wall and the price of a promise



Case study: $\lambda = 42$, $c = 3$, $h = 1.5 \Rightarrow \mu^* = 46.6$, $\rho = 90\%$, $w = 13$ min.

- Between $\rho = 90\%$ and 99% the wait multiplies by **ten**: a mathematical conflict, not an organizational one.
- Promise $w \leq w_{\max}$: from 12 to 9 min costs 1.85 €/h; from 3 to 2 min **28.95 €/h** ($dC/dw_{\max} = -c/w_{\max}^2 + h\lambda$).
- λ uncertain in $[36, 48]$: the nominal optimum **breaks down** ($\mu^* < 48$); the robust one ($\mu = 52.9$) costs +13%.

The newsvendor (stochastic LP)

News vendor — deciding before knowing

In words. A single quantity q , chosen *before* observing the demand D ; objective: minimum expected cost of overage (c_o) and underage (c_u).

Model and the quantile rule

$$\min_{q \geq 0} c_o \mathbb{E}[(q - D)^+] + c_u \mathbb{E}[(D - q)^+] \quad \implies \quad F(q^*) = \frac{c_u}{c_u + c_o}$$

Worked example ($D \in \{80, 100, 120\}$, $c_u = 9$, $c_o = 4$)

$C(80) = 180$, $C(100) = 86.7$, $C(120) = \mathbf{80}$: one orders the **maximum**, not the mean — $F(100) = 0.67 < \alpha^* = 0.692$.

Continuous case ($\mu = 100$, $\sigma = 20$, $c_u/c_o = 9/4$): $q^* = F^{-1}(0.6923) = 110 \neq 100$.

News vendor — scenario LP and value of the stochastic solution

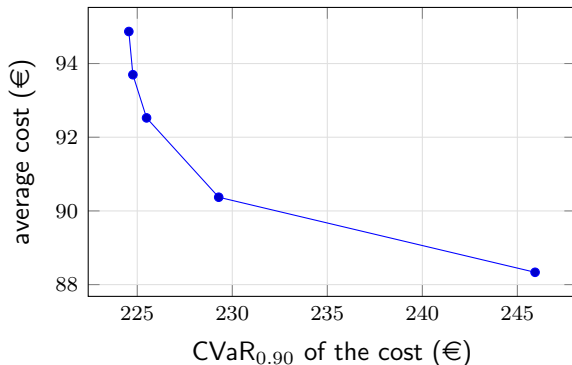
The positive-part trick (no binaries)

$$\min c_o \sum_{s=1}^k \pi_s o_s + c_u \sum_{s=1}^k \pi_s u_s \quad \text{subject to} \quad o_s \geq q - d_s, \quad u_s \geq d_s - q, \quad q, o_s, u_s \geq 0$$

At the optimum $o_s = (q - d_s)^+$ and $u_s = (d_s - q)^+$: the objective squeezes them.

- The q of the LP = the empirical quantile (108.6 with 600 scenarios): a theorem.
- **VSS** = 11.17 €/cycle (−11%): the gain from modelling uncertainty instead of ordering “the mean”.
- Stability: with 30 scenarios q moves by ± 3.5 units across replications — the scenarios are a sample.
- Service level: 95% no-stockout costs +45.59 €; the fill rate at q^* is already 96% — **read the SLA carefully.**

Newsvendor — risk aversion and multi-product



- Mean-CVaR: from $\lambda = 0$ to 1 one orders more ($108.6 \rightarrow 116.2$): $+6.5$ € of average buy -21.4 € of tail.
- Steep frontier at the start: much protection almost for free.
- Multi-product with a 1.200 € budget and correlated demands ($\rho = 0.7$): budget dual -0.55 (a 55% return!); with $\rho = 0$ the expected cost falls by 18%: correlation is a cost.

VaR and CVaR (LP)

VaR and CVaR — measuring the tail

L random loss, $\alpha \in (0,1)$:

$$\text{VaR}_\alpha(L) = \inf\{\eta : \mathbb{P}(L \leq \eta) \geq \alpha\},$$

$$\text{CVaR}_\alpha(L) = \text{average of the worst tail of mass } 1 - \alpha.$$

- CVaR is **convex** and subadditive (it rewards diversification); VaR is not.
- Worked example (losses $\{2.4, 5, 7, 12, 20\}$, $\alpha = 0.8$): $\text{VaR} = 12$; $\text{CVaR} = 12 + \frac{1}{0.2} \cdot \frac{1}{6}(20 - 12) = \mathbf{18.67}$; average = 8.33.
- Three numbers, three stories: the average reassures, the VaR gives a threshold, the CVaR says how much it hurts when things go badly.

The Rockafellar–Uryasev linear formulation

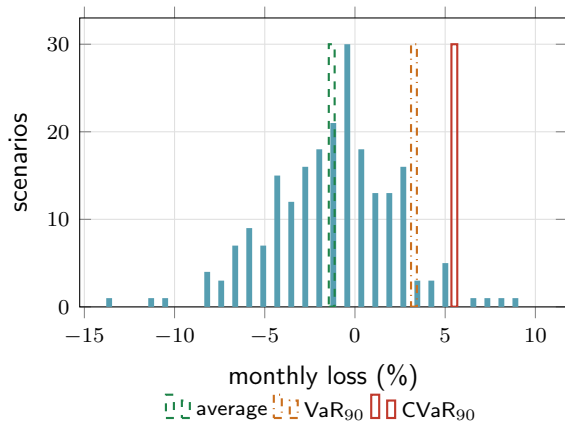
Scenarios $s \in \{1, 2, \dots, k\}$ with probabilities π_s ; linear loss $\ell_s(\mathbf{x})$; threshold η (a *free* variable) and excesses $\xi_s \geq 0$.

LP model

$$\begin{aligned} \min \quad & \eta + \frac{1}{1-\alpha} \sum_{s=1}^k \pi_s \xi_s \\ \text{subject to} \quad & \xi_s \geq \ell_s(\mathbf{x}) - \eta, & \forall s \in \{1, 2, \dots, k\}, \\ & \xi_s \geq 0, & \forall s \in \{1, 2, \dots, k\}. \end{aligned}$$

- At the optimum η^* is a VaR and the objective is the CVaR: **a single LP**, both measures.
- Same positive-part trick as the newsvendor: only the scenarios in the tail contribute.

VaR and CVaR — mean-CVaR portfolio vs Markowitz



220 scenarios with **fat tails** (Student t), required return 8%.

- Same average loss, but CVaR₉₀: mean-CVaR 4.89% vs Markowitz 5.31% (−8%).
- Variance penalizes above and below the average; CVaR looks **only where it hurts**.
- With $\alpha = 0.99$ and 220 scenarios the tail has 2–3 points: unstable estimates — scenarios beyond the quantile are needed.

VaR and CVaR — two-stage supply chain: the cost of resilience

First stage: capacity x_a reserved from the suppliers (F1 cheap but fragile, F2 expensive but reliable). Second stage (per scenario): flows and shortages.

λ	reserved F1	reserved F2	average cost	CVaR ₉₀
0 (neutral)	161.5	134.2	1.724	2.902
0.5	92.7	212.7	1.803	2.217
1 (protected)	68.5	236.9	1.906	2.139

Take-away

As risk aversion grows, capacity migrates towards the reliable supplier: +79 € of average cost buy –685 € of CVaR — the number needed in discussions on dual sourcing and reshoring.

Arbitrage and pricing (LP)

Arbitrage — the problem

- One-period market: today prices s_i^0 ; tomorrow one of m **states of the world**, asset i pays s_{ij}^1 ; asset 0 is risk free ($s_0^0 = 1$, it pays $R = 1 + r$ everywhere).
- **Arbitrage**: cash in today with no risk tomorrow (type A), or free today and never at a loss, with a chance of a gain (type B).
- Detecting it is an LP; duality is the theory of pricing.

The arbitrage detection LP

Free positions x_i (buying or short selling)

$$\begin{aligned} \min \quad & \sum_{i=0}^n s_i^0 x_i \\ \text{subject to} \quad & \sum_{i=0}^n s_{ij}^1 x_i \geq 0, & \forall j \in \{1, 2, \dots, m\}, \\ & x_i \geq 0, & \forall i \in \{0, 1, \dots, n\}. \end{aligned}$$

- The zero portfolio is feasible: the optimum is always ≤ 0 .
- If < 0 : type A and the model is **unbounded** (homogeneous constraints) $\Rightarrow$ normalization $\sum_i s_i^0 x_i \geq -1$ (“cash in at most 1 today”).

Duality is pricing

Every state gives a $p_j \geq 0$; every free position gives an **equality** dual constraint:

$$\sum_{j=1}^m s_{ij}^1 p_j = s_i^0 \quad \forall i \in \{0, 1, \dots, n\}, \quad p_j \geq 0.$$

From asset 0: $\sum_j p_j = 1/R \Rightarrow q_j = R p_j$ are **risk-neutral probabilities** and

$$s_i^0 = \frac{1}{R} \sum_{j=1}^m q_j s_{ij}^1 :$$

today's price is the discounted expected value of the payoffs under q .

The fundamental theorem of asset pricing

Theorem

No arbitrage (neither type A nor type B) $\iff$ there exists a risk-neutral measure with $q_j > 0$ for every state $j \in \{1, 2, \dots, m\}$.

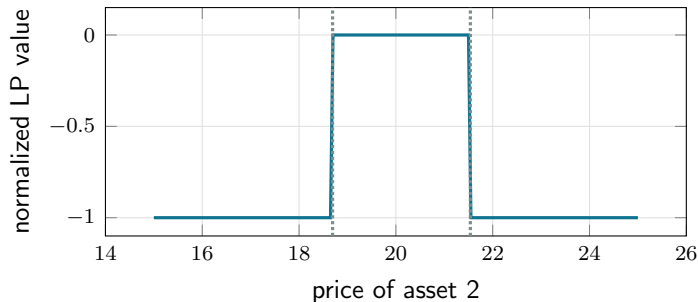
- No type A $\Rightarrow$ optimum 0 $\Rightarrow$ by **strong duality** the dual is feasible: a measure exists.
- No type B $\Rightarrow$ all the state constraints are binding $\Rightarrow$ by **complementary slackness** there is a measure with $q > 0$.
- Conversely, with $q > 0$ every riskless strategy costs ≥ 0 .

Arbitrage — case study

Three states, $R = 1.04$; payoff of asset 1 (10, 15, 13), asset 2 (30, 15, 25).

- **Prices (6, 20)**: UNBOUNDED; normalized: optimum -1 . By hand: $(-27, 1, 1)$ cashes in 1 today, payoff $(11.92; 1.92; 9.92) \geq 0$.
- **Prices (13, 18.69)**: optimum 0; from the duals of the state constraints $q = (0.296; 0.704; 0)$ — and the s_i^0 are reproduced exactly.
- **Pricing a call** (asset 1, strike 12, payoff $(0.3, 1)$): complete market $\Rightarrow$ unique price 2.0308; incomplete (assets 0 and 1 only) $\Rightarrow$ range $[1.4615, 2.0308]$.

Arbitrage — the range of consistent prices



- Two LPs (min/max over the consistent measures): $s_2^0 \in [18.69, 21.54]$; inside, value 0; outside, -1 : arbitrage in one direction or the other.

Support Vector Machine (QP)

SVM — the classifier as a QP

In words. Find the hyperplane that separates two classes maximizing the margin; the labels $y_i \in \{-1, +1\}$ are **data**, the variables ($\mathbf{w}$, b , slacks ξ_i) are all continuous.

Soft margin (primal)

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad y_i \left(\sum_{j=1}^p w_j x_{ij} + b \right) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

margin = $2/\|\mathbf{w}\|$; C = price of the violations.

Worked example (2 points): $\mathbf{w} = (0.5; 0.5)$, $b = -1$, margin $2\sqrt{2}$ = distance between the points; both are support vectors.

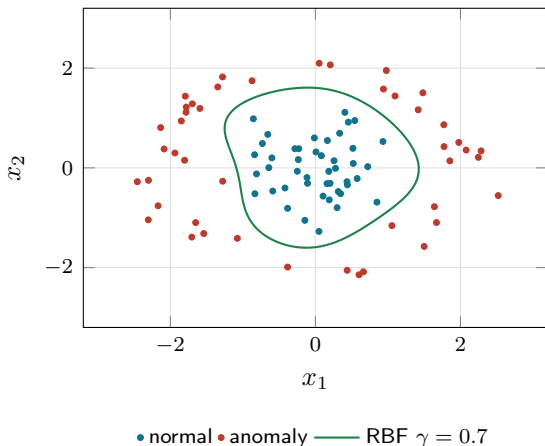
SVM — case study: credit risk

90 customers, two indicators; as C varies:

	$C = 0.05$	$C = 1$	$C = 20$
margin	2.42	1.28	0.36
training errors	2	2	0
points inside the margin	22	7	3

- **Dual:** $\max \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}'_i \mathbf{x}_j$, $0 \leq \alpha_i \leq C$, $\sum_i \alpha_i y_i = 0$: same w , b and value (strong duality, 4.7868).
- Complementarity: $\alpha_i = 0$ (83 points, irrelevant); $0 < \alpha_i < C$ (2, **on the margin**: they give b); $\alpha_i = C$ (5, inside the margin).
- Imbalanced classes: cost $10\times$ on the defaulters $\Rightarrow$ recall 100% at the price of precision 97%.

SVM — kernels and SVR



- The dual depends on the data only through $\mathbf{x}_i' \mathbf{x}_j$: replacing it with $K(\mathbf{x}, \mathbf{z}) = e^{-\gamma \|\mathbf{x} - \mathbf{z}\|^2}$ gives curved boundaries **while staying in a convex QP**.
- Small γ : smooth boundary; $\gamma = 5$: 61 support vectors out of 90, the model **memorises** (overfitting). C and γ by cross-validation.
- **SVR**: a $\pm \varepsilon$ tube with no penalty; on the price-demand data it recovers $-10.86p + 220.3$ (true: $-11p + 220$) using only the 14 points outside the tube.

Robust and quantile regression (LP)

Regression — estimating the parameters with an LP

In words. Choose the coefficients of a linear function that predicts demand; the objective pays τ for the deviations above and $1 - \tau$ for those below (u_i and v_i , the two parts of the deviation).

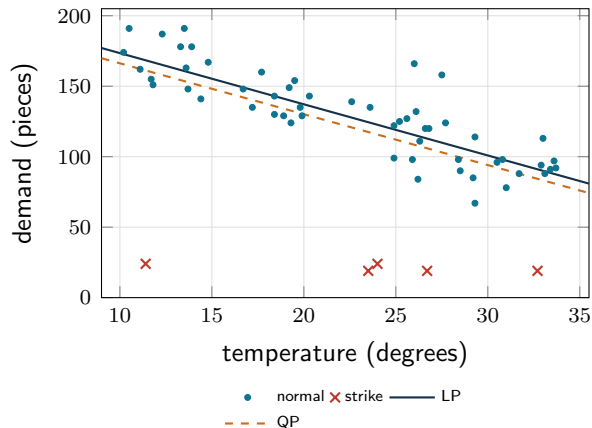
Quantile regression

$$\min_{w,b,u,v} \sum_{i=1}^n [\tau u_i + (1 - \tau)v_i] \quad \text{subject to} \quad \sum_{j=1}^p w_j x_{ij} + b + u_i - v_i = y_i, \quad u_i, v_i \geq 0$$

w_j and b free; $\tau = 1/2$ gives the median (absolute deviations), a large τ a prudent prediction on the high side.

With no features ($p = 0$) the optimum is the **empirical quantile**: with 88, 96, 104, 112, 120, 132, 148 and $\tau = 9/13$ we get $\tilde{b} = 120$ and $\tilde{z} = 680/13 = 52.31$ — it is the newsvendor quantile rule.

Regression — robustness and support points



- Five strike days out of 60: **least squares** pay the deviation squared and the intercept drops by 11 pieces; the **LP** stays where the normal data are ($209.77 - 3.63t$ against $213.58 - 3.72t$ of the QP cleaned by hand).
- **Duals:** $-(1 - \tau) \leq \tilde{\pi}_i \leq \tau$; above the estimate $\tilde{\pi}_i = \tau$, below $-(1 - \tau)$, and only the $p + 1$ **support points** have an interior dual — the support vectors of regression.
- $\sum_i \tilde{\pi}_i = 0$ (dual of the intercept) is the quantile property: 29 above, 29 below, 2 interpolated.
- Typical day: median 126.3, 9/13 quantile **132.1** against 148 for the unconditional quantile.

Lab organization

The four labs and the deliverable

Lab 1	Production and inventory	LP, duals, checking the shadow prices
Lab 2	Markowitz	QP, frontier, fragility of the estimates
Lab 3	Pricing <i>or</i> budget	NLP, concavity, numerical KKT
Lab 4	Project of your choice	managerial presentation

Report (max 8 pages): problem and assumptions → model (constraints explained) → data → results → sensitivity (full protocol) → **managerial recommendation in ten lines with no formulas.**

Grading: formulation 30% · implementation and verification 25% · sensitivity 25% · interpretation 20%.

The most common mistakes (checklist)

1. Reading `.X` or `.Pi` without checking `m.Status`.
2. Forgetting `lb=-GRB.INFINITY` on the free variables (b, η) .
3. Using a shadow price outside its validity range.
4. Getting the sign of the duals wrong in minimization problems.
5. Updating the RHS of a constraint with constants on the left.
6. Optimizing a single objective when there are two (pure minimax).
7. Choosing the hyperparameters by looking at the test set.
8. Six decimal places from estimates that wobble at the second.

Materials

- **Full lecture notes:** theory, examples worked out by hand, case studies, code, sensitivity, exercises (with solutions for instructors).
- **Python scripts:** one per chapter; `python/run_all.py` regenerates data, results and figures.
- **Solver guide:** `GUIDA_GUROBI.md`.
- **Data** in CSV with fixed seeds: everything is reproducible.

Enjoy the lab!